

Homework Assignment 2

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November 05, 2025

```
knitr::opts_chunk$set(fig.width=7, fig.height=5, echo = TRUE, message = FALSE, warning = FALSE)
options(digits = 4)
```

```
library(tidyverse)
library(patchwork)
```

```
# Read in the cleaned data
load('../data/biomarker-clean.RData')
```

```
# Get the variable names of the raw data for comparison
var_names <- read_csv('../data/biomarker-raw.csv',
                      col_names = F,
                      n_max = 2,
                      col_select = -(1:2)) %>%

  t() %>%
  as_tibble() %>%
  rename(name = V1,
         abbreviation = V2) %>%
  na.omit()
```

```
# Read in the raw data
biomarker_raw <- read_csv('../data/biomarker-raw.csv',
                          skip = 2,
                          col_select = -2L,
                          col_names = c('group',
                                         'empty',
                                         pull(var_names, abbreviation),
                                         'ados'),
                          na = c('-', '')) %>%
  filter(!is.na(group))
```

```
# Show the impact of outliers in raw data
# Pick a protein to visualize
protein <- "IRF1"
```

```
# Calculate stats for the protein
raw_stats <- biomarker_raw %>%
  filter(!is.na(!sym(protein))) %>%
  summarise(
    Mean = mean(!sym(protein)),
    Median = median(!sym(protein)),
```

```

    StdDev = sd(!sym(protein)),
    Min = min(!sym(protein)),
    Max = max(!sym(protein))
  )

cat(paste("--- Statistics for RAW", protein, "---\n"))

## --- Statistics for RAW IRF1 ---

print(raw_stats)

## # A tibble: 1 x 5
##   Mean Median StdDev   Min   Max
##   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 1179.  1101.   286.   658. 2866.

cat("\nNotice the large difference between the Mean and Median.\n")

##
## Notice the large difference between the Mean and Median.

cat("This indicates the data is heavily skewed by high-value outliers.\n")

## This indicates the data is heavily skewed by high-value outliers.

# Re-create the preprocessing steps

# Trim outliers
trim <- function(x, .at){
  x[abs(x) > .at] <- sign(x[abs(x) > .at])* .at
  return(x)
}

# Separate each step in data transformation
# Log10 transform
biomarker_log <- biomarker_raw %>%
  mutate(across(.cols = -c(group, ados),
    ~ log10(.x)))

# Log10 + Scale
biomarker_log_scaled <- biomarker_log %>%
  mutate(across(.cols = -c(group, ados),
    ~ scale(log10(.x))[, 1]))

# Log10 + Scale + Trim
biomarker_log_scaled_trimmed <- biomarker_log_scaled %>%
  mutate(across(.cols = -c(group, ados),
    ~ trim(.x, .at = 3)))

```

```

# Prepare data at different stages of preprocessing
df_raw <- biomarker_raw %>%
  select(group, all_of(protein)) %>%
  mutate(stage = "1. Raw")

df_log <- biomarker_log %>%
  select(group, all_of(protein)) %>%
  mutate(stage = "2. Log10")

df_scaled <- biomarker_log_scaled %>%
  select(group, all_of(protein)) %>%
  mutate(stage = "3. Log10 + Scaled")

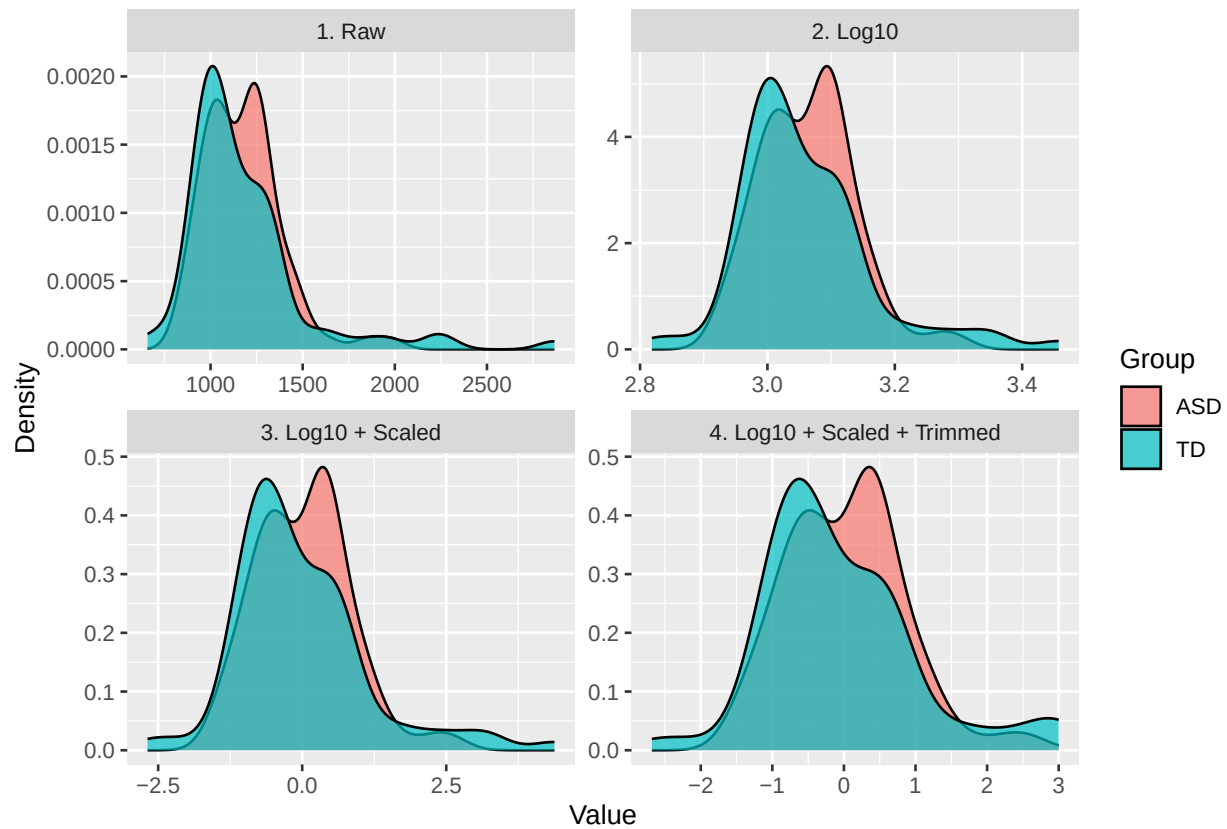
df_trimmed <- biomarker_log_scaled_trimmed %>%
  select(group, all_of(protein)) %>%
  mutate(stage = "4. Log10 + Scaled + Trimmed")

comparison_df <- bind_rows(df_raw, df_log, df_scaled, df_trimmed) %>%
  mutate(stage = factor(stage, levels = c("1. Raw",
                                           "2. Log10",
                                           "3. Log10 + Scaled",
                                           "4. Log10 + Scaled + Trimmed")))

# Plot the distributions at each stage
ggplot(comparison_df, aes(x = !!sym(protein), fill = group)) +
  geom_density(alpha = 0.7) +
  facet_wrap(~stage, scales = "free") +
  labs(
    title = paste("Distribution of", protein, "at Each Preprocessing Stage"),
    x = "Value",
    y = "Density",
    fill = "Group"
  )

```

Distribution of IRF1 at Each Preprocessing Stage



```
# Show the impact of the trimming function
df_before_trim <- biomarker_log_scaled %>%
  select(-group, -ados) %>%
  as.matrix()

df_after_trim <- biomarker_log_scaled_trimmed %>%
  select(-group, -ados) %>%
  as.matrix()

# Calculate total number of data points
total_points <- sum(!is.na(df_before_trim))

# Find how many points were outside the +/- 3 threshold
trimmed_points <- sum(abs(df_before_trim) > 3, na.rm = TRUE)
percent_trimmed <- (trimmed_points / total_points) * 100

# Print the findings (these are the results for your report)
cat("Total protein measurements:", total_points, "\n")
```

```
## Total protein measurements: 202818
```

```
cat("Observations trimmed (value > 3 or < -3):", trimmed_points, "\n")
```

```
## Observations trimmed (value > 3 or < -3): 2379
```



```
cat("Percentage of data trimmed:", round(percent_trimmed, 4), "%\n")

## Percentage of data trimmed: 1.173 %

# Visualize the impact of trimming
trim_df <- tibble(
  value_before_trim = as.vector(df_before_trim),
  value_after_trim = as.vector(df_after_trim)
) %>%
  filter(!is.na(value_before_trim))

# Scatterplot
ggplot(trim_df, aes(x = value_before_trim, y = value_after_trim)) +
  geom_point(alpha = 0.1) +
  # Add a reference line
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
  # Add lines showing the trimming threshold
  geom_vline(xintercept = c(-3, 3), color = "blue", linetype = "dotted") +
  geom_hline(yintercept = c(-3, 3), color = "blue", linetype = "dotted") +
  labs(
    title = "Impact of trimming on All Protein Data",
    x = "Value Before Trimming (Scaled)",
    y = "Value After Trimming (Scaled)"
  )
)
```

